

Kai-Jie Lin

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Education

National Yang Ming Chiao Tung University (NYCU)

B.S. in Computer Science; GPA: 3.83/4.3

Hsinchu, Taiwan

Sept. 2021 – Jun. 2025

University of Illinois Urbana-Champaign (UIUC)

Exchange Student in Computer Science

Illinois, USA

Aug. 2024 – Dec. 2024

Research Experience

NYCU Reinforcement Learning and Bandits Lab

Hsinchu, Taiwan

Research Assistant (Part-Time); Advisor: Prof. Ping-Chun Hsieh

Jan. 2025 – Present

- Designed a Q-Transformer framework for LLM that guides reasoning trajectories using long-term reward, optimizes latent trajectories via gradient ascent, and then decodes the latent representations into responses.
- Accelerated the training pipeline using FlashAttention for sampling stage and parallel target trajectory computation for training stage, reducing runtime from one week to eight hours.
- Integrated a Q-Transformer with Qwen-2.5-7B, improving accuracy for MMLU-Pro from 59% to 81% and cutting reasoning traces by 12% compared to Chain of Thoughts baseline; project aimed at *ICML'26* submission.

NVIDIA – NYCU Advanced Research Project

Hsinchu, Taiwan

Undergraduate Researcher; Advisor: Prof. Ping-Chun Hsieh

Jul. 2023 – Dec. 2024

- Investigated the hypervolume identifiability problem in Multi-Objective Bayesian Optimization (MOBO), showing that optimal policy learning requires history-dependent dynamics.
- Proposed BOFormer[*ICLR'25*], a Transformer-based Q-learning algorithm that learns non-Markovian policies via sequence modeling, overcoming myopic optimization behavior.
- Built pipelines to run 1K+ MOBO trials across 12 synthetic tasks and 6 real-world 3D reconstruction benchmarks, achieving up to 20% hypervolume improvement over SOTA baseline JES.

Work Experience

Rayleigh Vision Intelligence (RVI)

Hsinchu, Taiwan

AI Software Engineer

Feb. 2025 – Present

- Boosted Micro-LED wafer transfer efficiency by 4× through a Proximal Policy Optimization (PPO) algorithm benchmarked against existing greedy baseline.
- Built a horizontally sharded PostgreSQL database with hash partitions to manage 1B+ records, achieving <100ms query latency and 50K+ rows/second throughput using COPY-based inserts and chunked processing.
- Designed an Model Context Protocol (MCP) framework linking LLM to fab databases for wafer-transfer monitoring, reducing engineers' workload by 35% compared with the previous manual monitoring process.

Publications

BOFormer: A Generalized Deep Q-Learning Framework for Multi-Objective Bayesian Optimization

Yu-Heng Hung*, Kai-Jie Lin*, Yu-Heng Lin, Chien-Yi Wang, Cheng Sun, Ping-Chun Hsieh.

International Conference on Learning Representations (ICLR), 2025. (*Equal Contribution)

Selected Projects

Large Language Model Acceleration [Github]

NYCU, 2025

Course Project, Edge Artificial Intelligence

- Analyzed inference bottlenecks for quantized speculative decoding, revealing vLLM's Paged Attention achieved higher throughput than SGLang in speculative decoding by avoiding draft target contention.
- Accelerated Llama-3.2-3B model via General-Purpose Transformer Quantization (GPTQ) with vLLM inference and speculative decoding, boosting throughput from 3.90 toks/s to 91.97 toks/s at the same PPL 11.12.

GPU Kernel Optimizations [Github]

UIUC, 2024

Course Project, Applied Parallel Programming

- Optimized the convolution kernel by fusing unroll, GEMM and permutation kernels into a single CUDA kernel and integrating Tensor Core based GEMM with shared-memory tiling.
- Achieved a 3× speed-up, reducing two-layer convolution op-time from 200 ms to 65 ms with batch size 10K while preserving model accuracy.

Multispeaker Transcription [Github]

Hsinchu, 2023

Hackathon Project, Meichu Hackathon (Google Group)

- Developed an Android app and RESTful backend enabling real-time multi-speaker transcription for accessibility, using audio segmentation and SpeechBrain-based speaker diarization.
- Integrated PaLM-based post-processing for grammatical correction and style normalization, raising transcript readability by 25%; awarded Bronze Prize in the Google Accessibility Group category.

Honors & Awards

Bronze Medal, 2022 / 2023 ICPC Asia Taoyuan Regional Programming Contest